

Araz Hamayeli Mehrabani

Mechanical Engineer | Wind Turbine Simulation, Loads & Structural Dynamics

Wedel, Germany | +49 1575 436 3238 | araz.h.mehrabani@gmail.com

linkedin.com/in/araz-mehrabani-2467bab6 | arazmehrabani.github.io

Available for full-time employment from March 2027



Professional Profile

Mechanical engineer with nearly six years of industrial experience in product development, structural dynamics and simulation-supported design, now specialising in wind energy. Hands-on experience with OpenFAST aeroelastic simulations, IEC 61400-1 design load cases, Python workflow automation, structural-load post-processing and wind-farm planning in windPRO and QGIS. Combines current wind-turbine simulation expertise with practical experience developing dynamic machinery and renewable-energy systems through manufacturing and installation.

Core Technical Competencies

Wind turbine simulation	OpenFAST, aeroelastic modelling, IEC 61400-1 design load cases, turbine-response and structural-load analysis
Engineering automation	Python scripting, simulation case generation, batch execution, monitoring, data organisation and post-processing
Wind-farm planning	windPRO, QGIS, turbine layout, energy-yield and wake assessment, noise and environmental assessment
Structural dynamics	ANSYS Workbench, modal analysis, harmonic response, vibration and resonance assessment
Programming & data	Python, MATLAB/Simulink, Power BI, Excel
Mechanical engineering	CATIA V5, SOLIDWORKS, Siemens NX, design for manufacturing, drawings and bills of materials

Selected Wind Energy Experience

Master's Thesis - Rotor-Based Wind-Speed Estimation **Ongoing**

- Developing an OpenFAST-based method to estimate rotor-effective wind speed from measurable turbine-response signals, using the turbine rotor as a sensor.
- Automating simulation preparation, signal processing and estimator evaluation using Python.

20 MW Offshore Wind Turbine - Aeroelastic Load Analysis **University project**

- Conducted OpenFAST simulations for a 20 MW offshore monopile wind turbine across IEC 61400-1 design load cases using Python and MATLAB-supported workflows.
- Post-processed turbine-response channels and evaluated structural loads and dynamic behaviour under different environmental and operating conditions.
- Prepared engineering figures and comparative result sets to support interpretation of load-driving conditions and turbine response.

OpenFAST Simulation Automation **University project**

- Developed Python tools for OpenFAST case generation, batch execution, simulation monitoring and automated output organisation.

Wind-Farm Planning and Environmental Assessment **University project**

- Completed a site-based wind-farm planning study using windPRO and QGIS, including turbine layout, energy-yield estimation, wake assessment and noise calculations.
- Evaluated project constraints and environmental effects to compare planning alternatives and their influence on expected energy production.

Professional Experience

Development Engineer - Solar Tracking Systems & Automated Mechanisms

01/2022–02/2023

Paydar | Solar-Energy Systems

Tehran

- Designed photovoltaic tracking structures, mounting components and mechanical assemblies in SOLIDWORKS, supported by structural analysis in ANSYS.
- Developed an electric-motor-driven panel-cleaning mechanism with rail-and-wheel guidance and supported it through detailed design, manufacturing, assembly and successful on-site installation.
- Redesigned mechanical components to improve manufacturability, contributing to an approximately 10% reduction in production scrap and raw-material waste.
- Prepared manufacturing drawings, bills of materials and engineering documentation in SAP-based systems.

Mechanical Engineer - Machinery Development & Structural Dynamics

03/2017–12/2021

TehranRigzar | Mining & Mineral-Processing Equipment

Tehran

- Co-developed a high-frequency vibrating screen from laboratory studies and concept development through detailed design, manufacturing, assembly and customer installation.
- Performed modal and harmonic-response analyses in ANSYS to evaluate natural frequencies, resonance behaviour and dynamic response and to support structural design revisions.
- Supported the transition from a legacy crankshaft-driven screen to a vibration-motor-driven design with a simpler mechanical architecture.
- Calculated operating requirements with the engineering team and supported vibration-motor selection, manufacturing and assembly.

Education

M.Sc. Wind Energy Engineering - Current grade average: 1.9

03/2023–present

Flensburg University of Applied Sciences, Flensburg, Germany

Expected graduation: February 2027

M.Sc. Mechanical Engineering

09/2014–01/2017

Hakim Sabzevari University, Sabzevar

Control systems and robotics

Developed and implemented a MATLAB/Simulink motion-control system for precise movement and positioning of an autonomous underwater welding robot.

B.Sc. Mechanical Engineering

09/2008–06/2013

Islamic Azad University, Tehran

Mechanical design and vibration analysis

Designed a high-frequency screening machine in CATIA V5 and evaluated its structural and vibration behaviour using ANSYS.

Technical Tools

Wind energy	OpenFAST, windPRO, QGIS
Programming & data	Python, MATLAB/Simulink, Power BI, Excel
CAE	ANSYS Workbench, Abaqus
CAD	CATIA V5, SOLIDWORKS, Siemens NX
Engineering systems	SAP, MS Project, technical documentation

Selected Training & Certifications

Python programming	HarvardX
ANSYS	IEEE training
SOLIDWORKS	Technical and Vocational Training Organization
CATIA V5	Amirkabir University of Technology

Languages

English	IELTS overall score 7.0
German	B1 (self-assessed); preparing for certification